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Summary

I design, build and operate an MLOps platform on on-prem Kubernetes. Over six months at HUIINNO I codified the entire path — ingestion → lakehouse → transformation → lineage → distributed training → experiment / model registry — in **Pulumi**, as **sole owner of cluster construction** on a two-person platform team (93% of stack-code changes and 83% of the ETL repo's commits are mine). I remove human bottlenecks rather than add tools, and I put AI coding agents alongside me to raise what one engineer ships to team scale. Earlier: iOS → backend → frontend → cloud data pipelines, then led a data engineering team.

Key Results (measured 2026-09)

Led the design and implementation of the entire MLOps pipeline — ingestion through the model registry, with no gap left manual, codified into 12 Pulumi stacks: infrastructure changes land as PRs and a new environment is rebuilt by composing stacks. (1,204 commits, 678 merged PRs.)

Bare-metal Kubernetes built from the company's own hardware — no cloud: **10 nodes / 235 running pods / 23 internal services** in production, SSO across all 23 via Dex (GitHub OIDC) + oauth2-proxy, the same identity carried into Kubernetes RBAC.

Removed the single-person dependency in dataset preparation — "ask one data scientist and wait" became scheduled Airflow 3.x + dbt + Apache Iceberg runs with lineage in DataHub; researchers now query directly. (1,599 commits; the shared framework, a product pipeline, MIMIC-IV and the operational ones are all mine.)

A team's worth of platform in six months — most of this Kubernetes MLOps platform was written with AI coding agents alongside me, and it is still running in production. Not speed alone: 719 SDK unit tests, CI gates, 23 incident write-ups and 30 PRDs went in with it.

Earlier: Led the AWS EMR / PySpark art-market ETL pipeline (Eazel, team lead).

What I Do Now

Data Engineer, HUIINNO (Dec 2025 -). **I design and build the platform that makes research faster, and lay the MLOps pipeline on top of it piece by piece.**

On bare-metal Kubernetes I operate every layer: network, storage, catalog, lineage, orchestration, distributed training, identity.

I am attaching a reproducibility contract plus MLflow tracking and a model registry so research code can move onto KubeRay; the contract stamps version, content hash, code commit, environment lock and seed onto a dataset at run time.

A quantified survey showed results surviving while the data, code and environment behind them did not — that is where this design started.

I am establishing the split where researchers own domain ETL and the platform owns execution, lineage, verification and retries.

Stack

Platform: Kubernetes (kubeadm) · Pulumi · Cilium · SeaweedFS · Dex/OIDC · Prometheus/Grafana

Data: Airflow 3.x · dbt · Apache Iceberg · DataHub · PostgreSQL (pg_duckdb) · ClickHouse

ML: Ray/KubeRay · MLflow

Languages & other: Python · SQL · JavaScript · Swift · Shell · AWS/GCP

Certifications & Languages

Certifications: Engineer Information Processing (Nov 2017) · Craftsman Information Processing (Jul 2009) — issued by HRD Korea

Languages: Korean (native) · English OPIc AL · Japanese (conversational)

Full experience and project details are in a separate document ([resume_english.pdf](#)).